SARA (Smart Assistive Reading Aid)

Results, benchmarks and latency · 14 October 2026

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This document collects every measured result of the project in one place: how accurate each function is, how fast it runs on the Raspberry Pi 5 and on the laptop, and what is not measured. Each number comes from a saved result file or from the benchmark report written on the Pi on 13 October 2026. Where a number was measured on a hot, throttled board, or only on the laptop, it says so.

1. Accuracy of each function

Table 1: Accuracy on data the models were not trained on

Function	Test data	Result
Taka recognition	2,282 held-out pictures made like the training pictures	Precision 0.995, recall 0.996, mAP@0.5 0.995, mAP@0.5:0.95 0.849
Taka recognition	1,536 photographs by other people (Bangla Money)	94.1% right, 17 wrong (hand-held detector, 0.75); earlier detector 89.6%, 50 wrong (91.5% at 0.25)
Taka recognition, notes held in a hand	1,127 NSTU-BDTAKA close-ups, one per photo	47.5% right (earlier detector 11.8%); Pi 8-bit: 33.5% (earlier 8.7%)
Unknown notes (one Taka, not a class)	101 Bangla Money photos	Announced as a value: 36.6% (earlier 22.8%); Pi 8.9% (earlier 5.0%)
Several notes in one picture	900 pictures of 1, 2 or 3 test-split notes on new backgrounds	Spoken total right 97.0% (earlier detector 99.2%); never too many counted
Counterfeit check, 1 view / 6 views	208 test notes, usual split, 3 training runs	98.2 ± 0.7% / 98.7 ± 0.3% (71.2 ± 11.4% at 1 view without prefix training)
Counterfeit check on unseen prints	222 test notes whose counterfeit prints are not in training	Network alone about 90%; with the watermark check 94.4% (1 view), 95.0% (6 views)
Safe spoken verdict	88 test counterfeits	0 passed as "likely genuine"; the glass never says "counterfeit"
Safe spoken verdict, genuine notes confirmed	Close-up views / whole-note photos by other people	57.5% / 0.4–2%
Watermark check (MobileNetV2, in the glass)	197 unseen-print notes	92.9% (AUC 0.976); 92.6% on all 222 with the learned watermark finder
Watermark check, DMWR (47,818 parameters)	Same 197 notes, 3 training runs	91.5 ± 0.3%; not significantly different from MobileNetV2
UV fibre check, with the colour condition	111 real UV photos of genuine notes; simulated fibre-free printed sheets	85.6% of genuine photos accepted; 88.6% of held-out simulated sheets rejected. No real counterfeit tested

Function	Test data	Result
Text reading	Rendered benchmark, test set (432 pictures)	Letter error 2.3% (Bangla 3.0%, English 1.5%) with the new recogniser; 2.5% before it; 9.3% before this work
Text reading	1,000 real word photos (ICDAR 2019 MLT)	Letter error 28.7% Bangla, 50.6% Latin script with the recogniser trained on real words (39.2% and 62.5% before); nothing read on 12% and 33%
Text reading, picture enlarged to 880 instead of 1,200 pixels	Rendered test set	3.15% (full-precision detector), 3.12% (8-bit): worse, so not adopted
Object detection, YOLOv8s	All 5,000 COCO val2017 photos	mAP@0.5 0.551; 0.521 through the glass path (0.493 before a fix)
Object detection, YOLOv8n / YOLOv8m	Same	0.453 / 0.607
Emotion	RAF-DB official test, 3,068 faces	86.5% (macro-F1 0.795)
Handwritten Bangla letters (fall-back)	38,608 characters from 305 unseen writers	90.0%
Voice commands (wake word "glass")	12 held-out synthetic voices, 468 commands per condition; not people	Clean 100%; soft 99.8%; whispered 98.9%; street noise 10 / 5 / 0 / -5 / -10 dB: 98.5 / 93.8 / 82.5 / 57.7 / 39.5%; people talking 10 / 5 dB: 54.3 / 26.1%
Voice commands, false commands	360 ordinary sentences per condition, 14 conditions	13 of 5,040 (212 without the wake word); median response 0.86 s after the speech ends
Bangla voice commands (Whisper base, offline)	4 unseen Bangla voices, 15 phrases each	45% clean, 62% with noise; no wrong command; not deployed
Bangla voice (Piper), machine-listener check	6 sentences given to an online Bangla speech recogniser	2 exact, 2 with small differences, 2 with wrong words

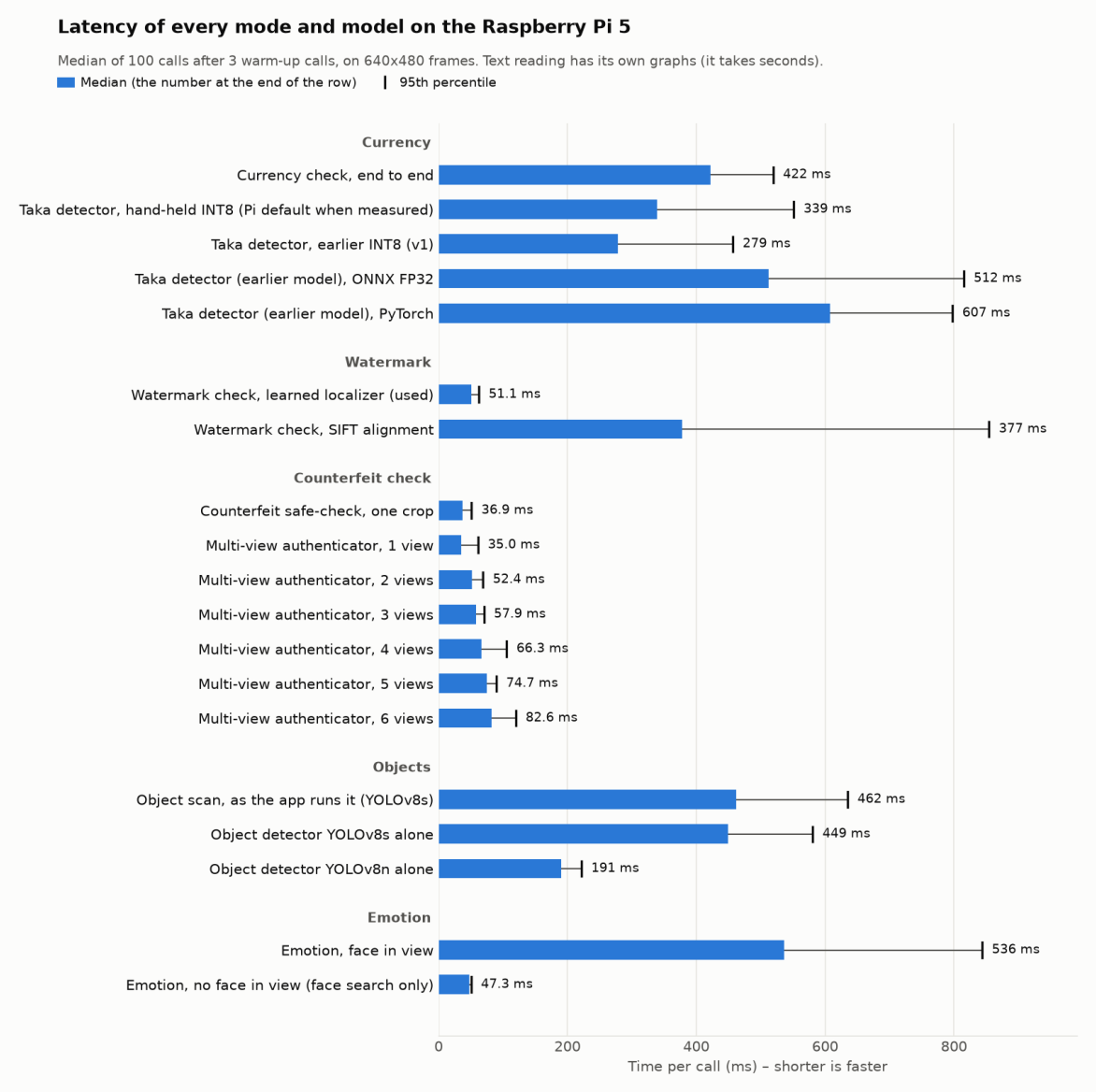
2. Latency on the Raspberry Pi 5

Measured on the glass's own Raspberry Pi 5 (8 GB, fan and heat sink, Logitech C270 webcam), 100 calls per row after 3 warm-up calls, from the Pi's benchmark report (benchmark_report/2026-10-29/REPORT.md). The currency check is given for the full-precision Taka detector the glass uses (benchmark_report/2026-10-29/raw/currency_current_code.json).

Table 2: Latency on the Raspberry Pi 5 (median; 95th percentile in brackets)

Part of the glass	Time
Currency check, end to end (full-precision detector, used)	532 ms (627 ms)
Taka detector alone: full precision (used) / 8-bit option	483 ms (566 ms) / 339 ms (551 ms)
Currency answer in use, press to speech (8 presses)	0.81 s
Watermark check, learned finder (used)	51 ms (63 ms)
Watermark check, SIFT alignment	377 ms (854 ms)
Counterfeit safe-check, one crop	37 ms (51 ms)

Part of the glass	Time
Counterfeit network, 1 view / 6 views	35 ms / 83 ms
Object scan as the app runs it (YOLOv8s)	462 ms (635 ms)
Object detector YOLOv8n alone	191 ms (222 ms)
Emotion, face in view	536 ms (844 ms)
Emotion, no face in view (face search only)	47 ms (51 ms)



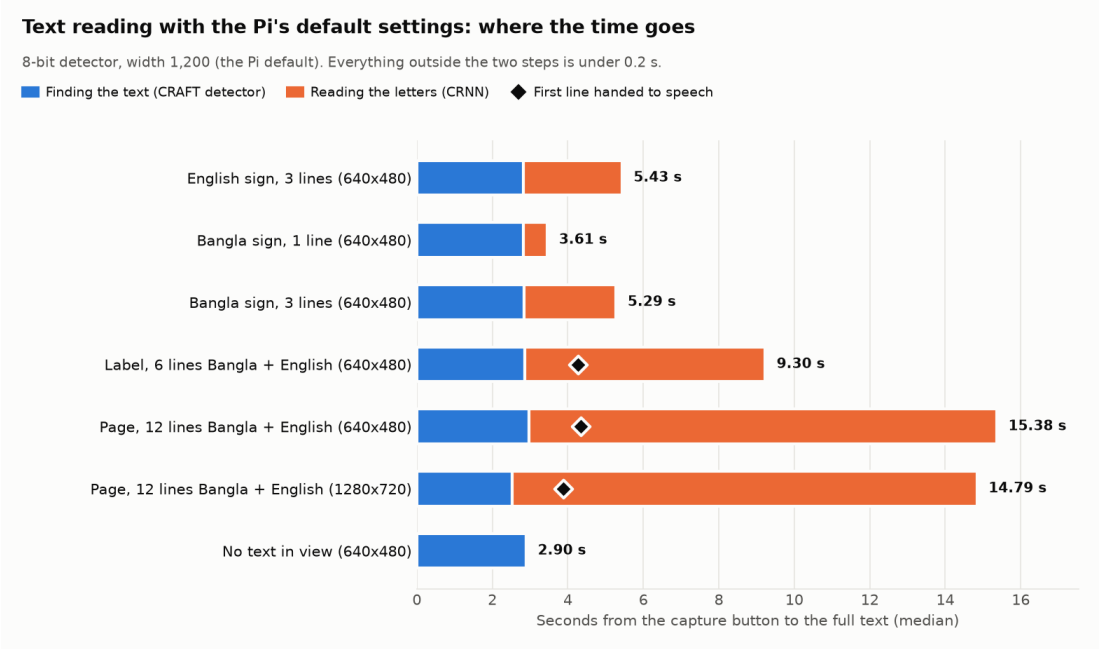
Latency of every mode and model on the Pi (from the Pi's benchmark report).

Text reading on the Pi

Table 3: Text reading on the Raspberry Pi 5 (8-bit text detector, read width 1,200; median of 4 or 5 captures)

Capture	Total	Finding the text	Reading the letters	First line spoken	Letter errors
English sign, 3 lines	5.43 s	2.83 s	2.61 s	–	0%
Bangla sign, 1 line	3.61 s	2.82 s	0.64 s	–	0%

Capture	Total	Finding the text	Reading the letters	First line spoken	Letter errors
Bangla sign, 3 lines	5.29 s	2.84 s	2.44 s	–	0%
Label, 6 lines, Bangla + English	9.30 s	2.87 s	6.35 s	4.27 s	0%
Page, 12 lines (640×480)	15.38 s	2.96 s	12.40 s	4.35 s	0%
Page, 12 lines (1280×720)	14.79 s	2.52 s	12.32 s	3.87 s	0%
No text in view	2.90 s	2.90 s	0 s	–	–



Text reading on the Pi: where the time of each capture goes (from the Pi's benchmark report).

- **One pass of the 8-bit text detector takes 2.5 to 3.0 s** on the Pi, against 9.9 to 11.2 s in full precision, and it made no letter error on these test pictures. On the rendered validation set it costs 0.57 points of letter error (3.15% against 2.58%), a stated exception to the project's 0.5-point rule, made for speed.
- **A picture with no text** answers "no text found" after one detector pass.
- **Real pages read with letter errors** through the camera, unlike these rendered test pictures; the field log of the app shows 12.2 to 22.2 s per printed page.
- **Steadier reading.** After a press the glass reads the sharpest frame of the next second: on validation text with simulated hand shake the letter error fell from 28.0% to 3.8% (savior_glass/results/ocr_bench/best_frame.json). *Real camera shake on the glass not measured.*

Other timings on the Pi

Table 4: Loading, speech and camera

What	Time
Loading the text reader (EasyOCR + 8-bit detector)	14.4 s
Currency mode, first switch into it / warm-up pass	2.6 s / 1.9 s

What	Time
Emotion detector / object mode / watermark checker / counterfeit network, loading	4.7 s / 1.6 s / 0.6 s / 0.3 s
Camera (C270), open to first frame	1.4 s
Camera, time between frames (640×480)	132 ms (7.6 frames a second)
Speech: making the audio for a short Bangla / English sentence	14.0 ms / 13.6 ms
Object scan and announcement in use, app log	0.74 s
Memory: whole benchmark process (peak) / running app with all modes	2.55 GB / 2.76 GB

Table 5: Board state

	Value
Temperature, processor sensor, 30 minutes of non-stop load	mean 82.8 °C, peak 87.0 °C
Soft temperature limit active during that run	55% of samples (fan at full speed)
Temperature of the board surface in normal use (thermometer, team)	57.8 to 62.3 °C
Board power: idle / sustained load / reading text	3.7 W / 7.8 W / about 9 W
Under-voltage	never



Temperature, clock and power during the benchmark (from the Pi's benchmark report).

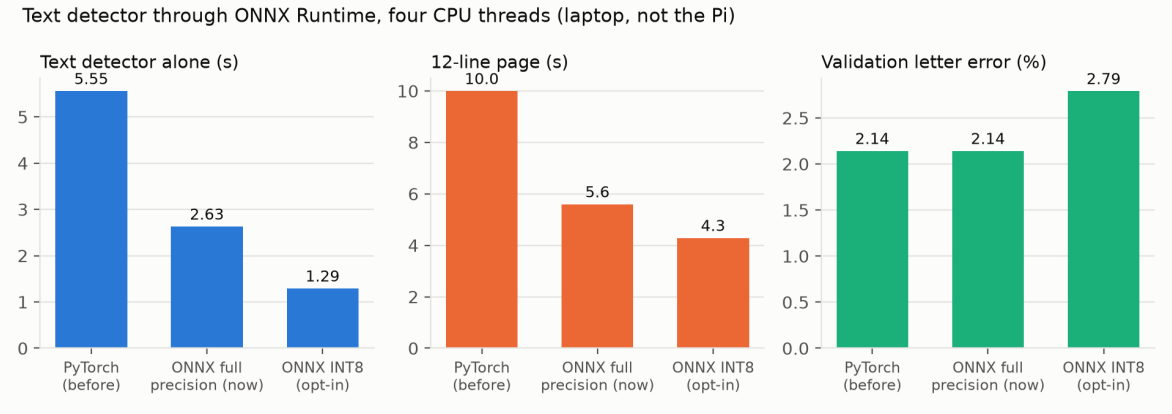
3. Speed measured on the laptop

These were measured on the development laptop (AMD Ryzen 7 5800H, RTX 3050), with the CPU limited to four threads where stated, the number of cores of the Pi. **They are not Pi numbers.** Only ratios can be expected to carry over.

Table 6: Laptop timings

What	Before	After	Accuracy
Text detector alone, 1,200×900 picture, 4 CPU threads (PyTorch → ONNX Runtime)	5.55 s	2.63 s	Identical output; validation error 2.14% both
12-line page, 4 CPU threads (same change)	9.99 s	5.59 s	Same
One benchmark picture, 4 CPU threads (same change)	2.30 s	1.22 s	Same
Text detector, 8-bit (INT8), 4 CPU threads	2.63 s	1.29 s (page 4.28 s)	Validation error +0.57 to +0.65 points
12-line page, beam search → greedy decoder (11 October run)	8.9 s	6.5 s	Validation error 2.18% → 2.14%

What	Before	After	Accuracy
12-line page spoken line by line: first line ready	nothing spoken until the page is done	about 5.5 s	Unchanged
Object scan through the glass path, 4 CPU threads (shrunk frame → frame as it is)	163 ms	173 ms	mAP@0.5 0.493 → 0.521
Speech: rendering one Bangla sentence of 1.5 to 4.3 s with the Piper voice (voice loaded)		0.2 to 0.5 s	Not timed on the Pi
Taka detector on the laptop GPU / CPU (PyTorch)		13.9 ms / 109 ms	
Taka detector, INT8 ONNX, laptop CPU		101 ms	
Counterfeit network, 6 views, laptop GPU		42.6 ms	
Watermark check, learned finder / SIFT, laptop CPU		15.9 ms / 97.9 ms	



Text detector through ONNX Runtime on four laptop CPU threads.

4. Results that corrected an earlier number

- **Objects.** 0.604 mAP came from 250 photos; on all 5,000 the same model scores 0.551.
- **Text on real photos.** 2.5% letter error on rendered text; 39% (Bangla) and 62% (Latin script) on real street words.
- **UV check.** The first design accepted 92.8% of genuine photos, and in a simulation also 93% of fibre-free printed sheets. With a colour condition: 85.6% of genuine photos accepted, about 89% of simulated sheets rejected.
- **Counterfeit benchmark.** About 98% on the usual split; about 90% on prints never seen in training (95.0% with the watermark).
- **Emotion on the Pi.** 48 ms was a frame without a face; with a face it is 536 ms.

5. Not measured

- The rebuilt voice on the Pi: how it sounds, and how long the Piper voices take to render there.
- Voice commands spoken by people, with the glass's microphone, in a room or a street.

- The UV check and the watermark check through the C270 webcam (their thresholds come from phone and JaalTaka photos).
- Any counterfeit note under the UV lamp; any function through the glass camera in daylight.
- The online description mode (the account has no credit).
- Battery life.
- A study with visually impaired users (trial sessions have started; no result is reported).

Sources. Pi figures: benchmark_report/2026-10-29/REPORT.md and its raw files. Laptop and accuracy figures: the result files under savior_glass/results/ and paper_evidence/.